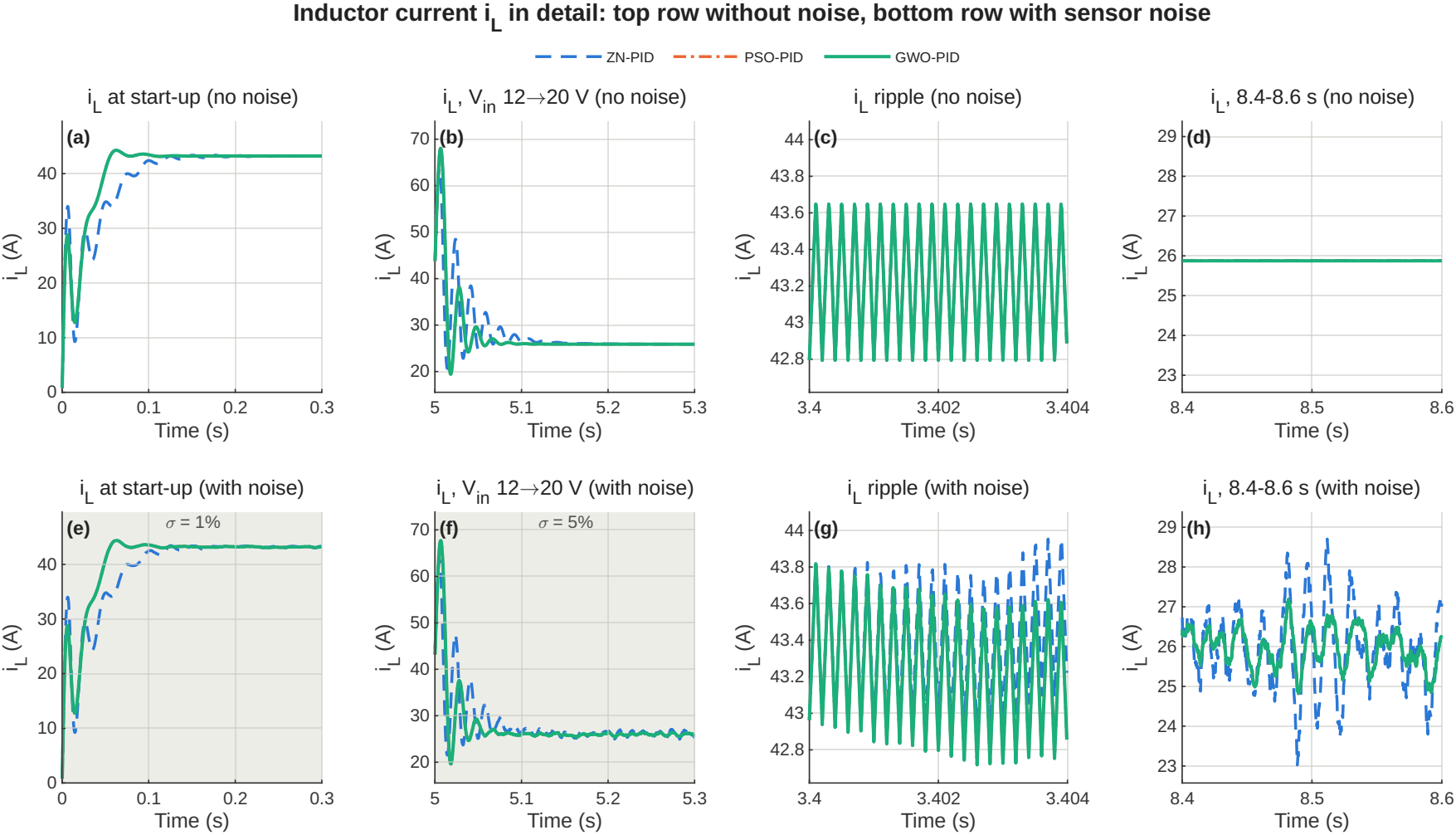

Inductor current i_L in detail: top row without noise, bottom row with sensor noise
ZN-PID
PSO-PID
GWO-PID
i_L at start-up (no noise)
i_L, V_in 12→20 V (no noise)
i_L ripple (no noise)
i_L, 8.4-8.6 s (no noise)
i_L at start-up (with noise)
i_L, V_in 12→20 V (with noise)
i_L ripple (with noise)
i_L, 8.4-8.6 s (with noise)
(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
σ = 1%
σ = 5%
i_L (A)
Time (s)